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Complementary subspaces

Let V be an inner product space. Let U be a non-empty subset of V. Then

$$U^{\perp} := \{ v \in V \mid \langle v, u \rangle = 0 \text{ for all } u \in U \}.$$

① $U^{\perp}$ is a subspace of V.

(i) $U^{\perp} \neq \phi$ $\because$ $\langle 0, u \rangle = 0$ for all $u \in U$
$\Rightarrow 0 \in U^{\perp}$

(ii) If $u_1, u_2 \in U^{\perp}$, then $u_1 + u_2 \in U^{\perp}$

$\Downarrow$

$\langle u_1, u \rangle = 0 \quad \forall u \in U$

$\langle u_2, u \rangle = 0 \quad \forall u \in U$

T.P. $\langle u_1 + u_2, u \rangle = 0$

$$\langle u_1 + u_2, u \rangle = \langle u_1, u \rangle + \langle u_2, u \rangle = 0 + 0 = 0 \quad \forall u \in U.$$

(iii) Let $u_1 \in U^{\perp}$, T.P. $\alpha u_1 \in U^{\perp} \quad \forall \alpha \in F$

$\Downarrow$

$\langle u_1, u \rangle = 0 \quad \forall u \in U$

$$\langle \alpha u_1, u \rangle = \alpha \langle u_1, u \rangle = \alpha . 0 = 0$$
$$\Rightarrow \alpha u_1 \in U^{\perp} \quad \forall \alpha \in F.$$

question ① $\{0\}^{\perp} = V$ ② $V^{\perp} = \{0\}$.

Let $0 \neq v \in V^{\perp}$, $\langle v, v' \rangle = 0 \quad \forall v' \in V$

In particular $v' = v$

$\Rightarrow \langle v, v \rangle = 0 \Rightarrow v = 0$

③ Let U be a non-empty subset of an Inner product space V, then

$$U \cap V^{\perp} \subset \{0\}.$$

$U \cap U^{\perp} = \phi$ then we are done.

$U \cap U^{\perp} = \{0\}$.

Sol:- If $U \cap U^{\perp} = \phi$, then we are done.

If $U \cap U^{\perp} \neq \phi$, then T.P. $U \cap U^{\perp} = \{0\}$.

Let $u \in U \cap U^{\perp}$

$u \in U$ & $u \in U^{\perp}$

$\longrightarrow \langle u, v \rangle = 0 \quad \forall\ v \in U$

In particular $v = u$

we get $\langle u, u \rangle = 0 \Rightarrow u = 0. \Rightarrow U \cap U^{\perp} = \{0\}$.

positive definite

- $V \rightarrow$ Inner product space. Let U and W be two non-empty subsets of V.

If $U \subset W$, then $W^{\perp} \subset U^{\perp}$

Given $U \subset W$ T.P. $W^{\perp} \subset U^{\perp}$

Let $v \in W^{\perp}$

$\Rightarrow \langle v, w \rangle = 0 \quad \forall\ w \in W$

$\because U \subset W$

$\Rightarrow \langle v, u \rangle = 0 \quad \forall\ u \in U$

$\Rightarrow v \in U^{\perp}$

Thm:- Let V be an inner product space. Let U be a finite dimensional subspace of V. Then

$$V = U \oplus U^{\perp}.$$

$\Downarrow$

[① $V = U + U^{\perp}$], ② $U \cap U^{\perp} = \{0\}$.

we have shown that

$\ni\ u \in U,\ u' \in U^{\perp}$

(i) $V = U + U^\perp$

$\forall$ $v \in V$, $\exists$ $u \in U$, $u' \in U^\perp$

s.t. $v = u + u'$

, (ii) $U \cap U^\perp = \{0\}$.

we have shown that

$U \cap U^\perp \subseteq \{0\}$

$\because$ U and $U^\perp$, both are subspace $\Rightarrow$ $U \cap U^\perp = \{0\}$

$\because$ U is f.d. inner product space, using Gram Schmidt orthogonalisation, we have an orthonormal Basis of U.

Let $\{e_1, e_2, -, e_k\}$ be an orthonormal Basis of U.

Let $v \in V$

$$v = \underbrace{\langle v, e_1\rangle e_1 + \langle v, e_2\rangle e_2 + \text{—} + \langle v, e_k\rangle e_k}_{\in U} + \underbrace{v - \langle v, e_1\rangle e_1 - \langle v, e_2\rangle e_2 - \text{—} - \langle v, e_k\rangle e_k}_{\text{T.P. } \in U^\perp}$$

Denote $w = v - \langle v, e_1\rangle e_1 - \langle v, e_2\rangle e_2 - \text{—} - \langle v, e_k\rangle e_k$

T.P. $w \in U^\perp$

$\langle u, w\rangle = 0$ $\forall$ $u \in U$.

$\because$ $\{e_1, e_2, -, e_k\}$ is a Basis of U

$u = \alpha_1 e_1 + \alpha_2 e_2 + \text{—} + \alpha_k e_k$

$\langle u, w\rangle = \langle \alpha_1 e_1 + \alpha_2 e_2 + \text{—} + \alpha_k e_k, w\rangle$

$= \alpha_1 \underbrace{\langle e_1, w\rangle}_{=0} + \alpha_2 \underbrace{\langle e_2, w\rangle}_{=0} + \text{—} + \alpha_k \underbrace{\langle e_k, w\rangle}_{=0}$

If we prove that $\boxed{\langle e_i, w\rangle = 0 \quad \forall \quad 1 \le i \le k}$

$\langle u, w\rangle = 0$

If we prove that $\boxed{\langle e_i, w\rangle = 0 \quad \forall \quad 1 \le i \le k}$

then we get $\langle u, w\rangle = 0$.

T.P $\langle w, e_i\rangle = 0 \quad \forall \; 1 \le i \le k$

$$w = v - \langle v, e_1\rangle e_1 - \langle v, e_2\rangle e_2 - \_\_\_ - \langle v, e_k\rangle e_k$$

$$\langle w, e_1\rangle = \langle v - \langle v, e_1\rangle e_1 - \langle v, e_2\rangle e_2 - \_\_\_ - \langle v, e_k\rangle e_k, e_1\rangle$$

$$= \langle v, e_1\rangle - \langle v, e_1\rangle \underbrace{\boxed{\langle e_1, e_1\rangle}}_{1} - \langle v, e_2\rangle \overbrace{\langle e_2, e_1\rangle}^{=0} - \_\_\_ - \langle v, e_k\rangle \overbrace{\langle e_k, e_1\rangle}^{=0}$$

$$= \langle v, e_1\rangle - \langle v, e_1\rangle = 0$$

Similarly $\langle w, e_2\rangle = 0$, $\langle w, e_3\rangle = 0$ ___ $\langle w, e_k\rangle = 0$

$\Rightarrow w \in U^\perp$.

$\Rightarrow V = U + U^\perp$ & $U \cap U^\perp = \{0\} \Rightarrow V = U \oplus U^\perp$

* $V \to$ Inner product space, $U \to$ f.d. Subspace

$$(U^\perp)^\perp = U.$$

T.P. (i) $U \subset (U^\perp)^\perp$

Let $u \in U$

$\Rightarrow \langle u, v\rangle = 0 \quad \forall \; v \in U^\perp$

$\Rightarrow u \in (U^\perp)^\perp$.

(ii) $(U^\perp)^\perp \subset U$. Let $v \in (U^\perp)^\perp$, $v \in V$

$V = U \oplus U^\perp$

$v = u + w$, $u \in U$, $w \in U^\perp$.

$V = U \oplus U$

$v = u + w, \quad u \in U, \quad w \in U^{\perp}.$

- $v \in (U^{\perp})^{\perp}, \quad w \in U^{\perp} \Rightarrow \langle v, w \rangle = 0$
- $u \in U, \quad w \in U^{\perp} \Rightarrow \langle u, w \rangle = 0$

$$0 = \langle v, w \rangle = \langle u + w, w \rangle = \langle u, w \rangle + \langle w, w \rangle = 0 + \langle w, w \rangle$$

$$\Rightarrow \langle w, w \rangle = 0 \Rightarrow \underline{w = 0}$$

$$\Rightarrow v = u \in U, \quad (U^{\perp})^{\perp} \subset U.$$

- $C_{\mathbb{R}}[-1,1] \longrightarrow$ set of cont. real valued function defined on $[-1,1]$

Check! $C_{\mathbb{R}}[-1,1]$ is a vector space over $\mathbb{R}$.

$f, g \in C_{\mathbb{R}}[-1,1] \qquad f+g \in C_{\mathbb{R}}[-1,1], \quad \alpha f \in C_{\mathbb{R}}[-1,1]$

Define map

$$\langle f(x), g(x) \rangle = \int_{-1}^{1} f(x)\, g(x)\, dx$$

Check! $\langle f(x), g(x) \rangle$ is an inner product over $C_{\mathbb{R}}[-1,1]$

- linear in first co.
- ~~linear in second con~~
- symmetry
- positive definite

$f(x) = x$

$f(x) = x^2$

$f(x) = x^2 + x$

$$U = \{ f(x) \in C_{\mathbb{R}}[-1,1] \mid f(0) = 0 \}$$

Check! Is this subspace of $C_{\mathbb{R}}[-1,1]$?!

$$U^\perp = \{ g(x) \in C_R[-1,1] \mid \langle g(x), f(x) \rangle = 0 \ \forall f(x) \in U \}.$$

$\int_{-1}^{1} g(x) f(x)\, dx = 0$, $f(x) \in U \to f(0) = 0$

$U^\perp = \{0\}$.

- $V = U \oplus U^\perp$

$\overset{V}{C_R[-1,1]} \neq \underset{f(0)=0}{U} \oplus \underset{0}{U^\perp}$

$\because$ U is Not f.d. subspace.

$(U^\perp)^\perp \neq U$. In this case

* U is f.d subspace, $V \to$ inner product space

$$V = U \oplus U^\perp$$

$V \to$ f.d. inner product space,

$$\dim V = \dim U + \dim U^\perp$$

* $\dim(U_1 + U_2) = \dim U_1 + \dim U_2 - \dim(U_1 \cap U_2)$

Take $U_1 = U$, $U_2 = U^\perp$, $U \cap U^\perp = \{0\}$